

Bistable paper origami

Design development and experimental evidence

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23 September 2026

The goal was to fabricate paper Kresling and Miura structures with distinct configurations that remain after release. The final prototypes demonstrate a taller/lower Kresling cycle and local Miura pop-through. Design screening guided fabrication; the videos supply qualitative physical evidence. The changes below record practical constraints and unsuccessful numerical checks without treating every discarded design as a failed physical experiment.

01 Kresling: seam geometry and fabrication constraints

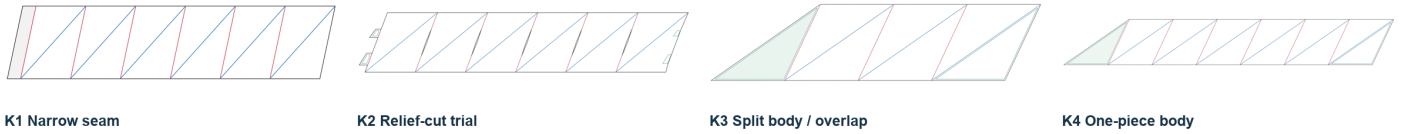


Figure 1. Saved Kresling patterns, at different display scales. K3 shows one of two body pieces. Red: mountain; blue: valley; black: cut. Cylinder geometry draws on [1].

K1 / Narrow seam. The full-height join interfered with collapse. The seam required re-design.

K2 / Relief-cut trial. Diamond openings and short tabs were impractical with scissors. The next search required no internal cuts.

K3 / Full overlap. A duplicate triangle replaced small tabs. The 60 mm layout needed two pieces and two body joins.

K4 / One piece. Scaling to 43 mm rim edges and rotating the print by 30° fitted one body on Letter; caps print separately.

02 Miura: tessellation complexity and facet modification

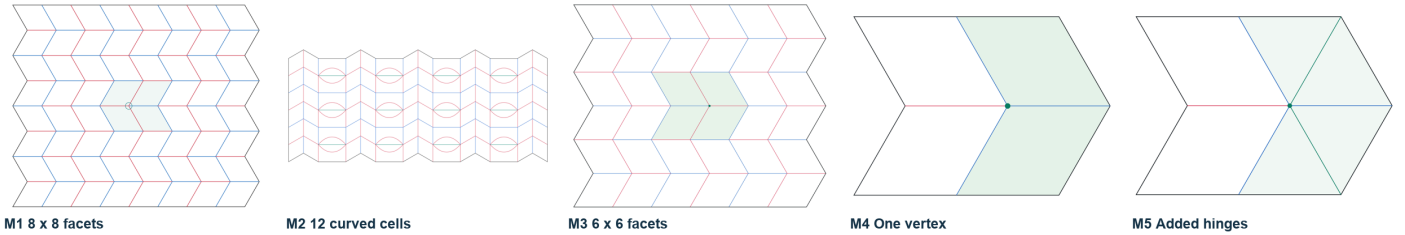


Figure 2. Miura development: large straight sheet, curved 12-cell lens-box, smaller straight sheet, one vertex, and added facet hinges. Project adaptations of [2]–[4]; these are crease drawings, not simulation results.

- M1** 8×8 facets: many coordinated folds; direct numerical loading returned to the original state even though a separately seeded defect survived.
- M2** 4×3 curved cells: curved scoring and bonded connectors were difficult by hand. Full-panel press/release did not switch; a refined diagnostic patch did not complete a clean reverse transition.
- M3** 6×6 straight facets: a coarse second minimum vanished on refinement under the same assumptions. Other thin-sheet minima survived, but the finer switching solve did not converge.
- M4** 2×2 facets: the same 45 mm outline was easier to manipulate. An early model returned to A; revised stiffness assumptions gave a conditional cycle. The first physical clip was inconclusive.
- M5** Added diagonals through B and D: the 65 lb cardstock specimen exhibited A–B–A pop-through and retained both released configurations.

03 Final dimensions and fabrication

Kresling: $N = 6$; rim edge $a = 43$ mm; flat rise/shift 44.410/20.898 mm; valley/mountain angles to the rim $\alpha = 34.8^\circ$, $\beta = 64.8^\circ$. One full-triangle overlap and six flaps per cap.

Miura: original rhombus sides 45 mm; sectors $60^\circ/120^\circ$; flat outline 112.5×77.94 mm; three original valleys, one mountain and two added 45 mm diagonals. Six facet regions; 65 lb cardstock; no glue or holes.

Full-size templates: print at 100% on US Letter and check the 50 mm bar. Flat strip rise is not cylinder height. Exact stock for the filmed Kresling was not independently recorded.

Release evidence and mechanics

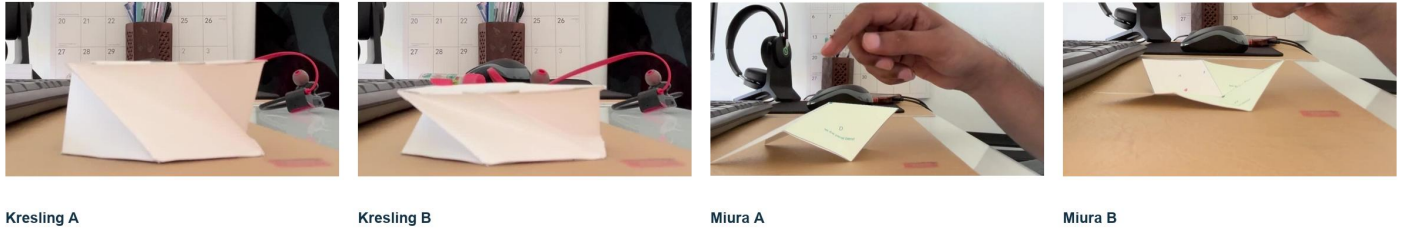


Figure 3. Frames from the videos, cropped for comparison. Both configurations are shown without hand contact.

04 Experimental observations

Kresling. Compression couples with twist. The sheet rebounds to a lower resting configuration, then reopens to a taller one. The HUD measures only normalized visible-outline height $h/h_{\max}$ during hands-free intervals, not physical millimeters. The held flat pose is excluded.

Miura. The edit shows two pop-through transitions and one A–B–A cycle. Uninterrupted retained shots show approximately 3.1 s in B and 3.9 s in returned A without contact. The piece is turned over between shots; timers restart after cuts and handling. Both edits are silent and preserve real-speed motion.

05 Mechanical role of auxiliary facet creases

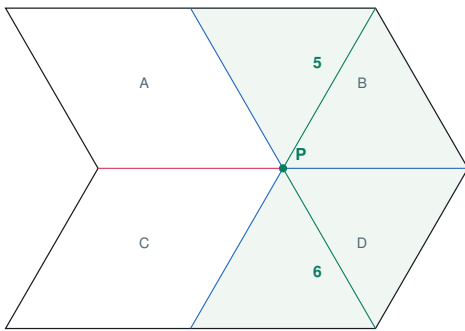


Figure 4. Green hinges connect P to opposite corners of B and D. They divide two rhombi into four triangles, giving six facet regions in total.

Ordinary Miura folding is a coordinated geometric motion, not itself a proof of two energy minima. [4] Silverberg’s pop-through model allows adjacent facets to bend and represents that deformation with two additional symmetric hinges, flat at rest. [2]

The permanent B/D creases give this specimen defined bending lines. They can lower resistance to inversion and change fold memory, allowing the surrounding panels to reconfigure while retaining stiffness away from those lines. This is a plausible explanation for the physical improvement, not proof that auxiliary creases are universally necessary.

An illustrative energy model is

$$U(\mathbf{q}) = \frac{1}{2} \sum_i k_i [\theta_i(\mathbf{q}) - \theta_{i,0}]^2 + U_{\text{panel}}(\mathbf{q}).$$

Nonlinear geometric compatibility can make U nonconvex even with simple crease springs. [5] A sufficient local test is $\nabla U = 0$ with a positive-definite constrained Hessian at each state; bistability also requires an intervening energy barrier. Here k_i is rotational stiffness; $\theta_{i,0}$ is the rest angle. Added folding freedom alone does not guarantee bistability.

Scope. The final added-crease specimen was not the exact system used in earlier simulations. Crease stiffness, force barriers and endurance were not measured. Kresling released heights of 32.68/9.83 mm are model predictions, not measured dimensions.

06 References and separate deliverables

- [1] Pagano, A., Leung, B., Chien, B., Yan, T., Wissa, A., & Tawfik, S. (2016). Multi-Stable Origami Structure for Crawling Locomotion. *ASME SMASIS*, V002T06A005. DOI: 10.1115/SMASIS2016-9071.
- [2] Silverberg, J. L., et al. (2014). Using origami design principles to fold reprogrammable mechanical metamaterials. *Science*, 345, 647–650. DOI: 10.1126/science.1252876.
- [3] Mirzajanzadeh, M., & Pasini, D. (2025). Reprogrammable curved-straight origami: Multimorphability and volumetric tunability. *Science Advances*, 11, eadu4678. DOI: 10.1126/sciadv.adu4678.
- [4] Schenk, M., & Guest, S. D. (2013). Geometry of Miura-folded metamaterials. *PNAS*, 110, 3276–3281. DOI: 10.1073/pnas.1217998110.
- [5] Waitukaitis, S., Menaut, R., Chen, B. G., & van Hecke, M. (2015). Origami Multistability: From Single Vertices to Metasheets. *Physical Review Letters*, 114, 055503. DOI: 10.1103/PhysRevLett.114.055503. Linked PDF: arXiv manuscript.

Project site, videos and full-size templates

ar3116-sketch.github.io/bistable-paper-origami/